

SINDH UNIVERSITY (PET) ENTRY TEST

Master Prep Notes

Topic Breakdown, Yearly Analysis & Most-Repeated Questions

Based on: Sindh University Past Papers (2013–2019) and PET-2025 Sample Paper

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1. Exam Structure & Weightage

Section	Weight	Covers
English	30%	Synonyms/Antonyms, Spelling, Voice, Narration, Tenses, Prepositions
General Science	20%	Basic Science, Physics, Chemistry, Computer
General Knowledge	20%	World Knowledge, Pakistan Studies, Islamiyat
Simple Arithmetic	20%	Arithmetic, Fractions, Percentage, Average
Intelligence Questions (IQ)	10%	Word Problems, Shape Problems

2. Master Topic List (By Section)

English (30%)

- Synonyms (word meaning matching)
- Antonyms (opposite meaning)
- Correct spelling identification
- Prepositions (fill in the blank)
- Active/Passive voice conversion
- Direct/Indirect (narration) conversion
- Sentence completion / fill in the blanks (tenses, modals, articles)
- Error identification (grammar errors in sentences)
- Choosing correct sentence structure
- Basic grammar concepts (pronouns, relative pronouns, parts of speech)

General Science (20%)

- Basic Physics (units, forces, waves, light, motion, Newton's laws, electricity)
- Basic Chemistry (chemical formulas, elements, symbols, periodic table)
- Biology (human body systems, cells, diseases, vitamins, plants)
- Computer basics (hardware/software, CPU, microprocessor)
- Scientific instruments and their uses (barometer, altimeter, sphygmomanometer, etc.)
- Astronomy (solar system, planets)

General Knowledge (20%)

- Pakistan Studies (history, geography, landmarks, barrages, rivers, mountains, cities)
- Islamiyat (Quran, Hajj, Prophet's life, Islamic history, Surahs)
- World Geography (capitals, currencies, rivers, mountains, oceans)
- Notable personalities (local and international)
- Static GK (organizations, awards, historic events)
- Sindh-specific knowledge (Sufi shrines, local culture, Shah Abdul Latif Bhittai)

Simple Arithmetic (20%)

- Basic arithmetic operations (multiplication, division, decimals)
- Fractions and percentages
- Ratios and proportions
- Averages
- Algebra (basic equations, exponents, expressions, factorization)
- Sets (union, intersection, subsets, De Morgan's Law)
- Basic geometry (area, perimeter, squares)
- Trigonometry basics (sin, cos values)
- Number systems (rational/irrational numbers, binary)
- Word problems (profit/loss, speed/distance, partnerships)

Intelligence Questions / IQ (10%)

- Number series completion
- Pattern recognition (shapes, figures)
- Logical reasoning puzzles
- Coding-decoding
- Direction-based problems

- Analogy-based problems
- Odd one out
- Simple word/letter puzzles

3. Year-by-Year Topic Breakdown

2013

English: Synonyms, fill-in-the-blank (prepositions/articles), parts of speech, active/passive voice, indirect narration, conditional sentences, correct sentence structure

General Knowledge: UN membership dates, historical figures (Amir Khusro), national languages/anthems, world capitals, G-8 members, explorers, world events, Islamiyat (Hajj, Azan/Namaz, Surahs)

General Science: Biology basics, physics (waves, mass), chemistry (formulas, atomic weight), DNA/genetics, periodic table

Arithmetic: Decimals, order of operations (BODMAS), algebra (exponents, expressions), sets, trigonometry basics

IQ: Number patterns, coded logic (letter/number substitution), distance/direction problems, classification (odd one out)

2014

English: Synonyms, antonyms, correct spellings, fill-in-the-blanks, indirect narration, passive voice

General Knowledge: Islamic calendar, Islamic history/battles, Pakistani leaders, landmarks (Badshahi Mosque, Hunza Valley), world buildings, Nobel Prize, Nishan-e-Haider

General Science: Chemical formulas, biology (frog anatomy, digestion), water purification, ozone layer, elements/gases, Newton's laws

Arithmetic: Multiplication/division, roots, ratio & partnership, profit/loss, algebra, sets, percentages, perimeter/area

IQ: Trick/logic questions, handshake problems, classification, number series, direction problems, word puzzles

2015

English: Synonyms, antonyms, prepositions, sentence completion, spellings, indirect narration, passive voice, error identification

General Knowledge: Quran facts, Pakistani PMs, Lahore Resolution, Sindh culture (Sasui-Punhu, Lal Shahbaz Qalandar), world geography, historical figures

General Science: Vitamins, human body systems, measuring instruments, gases, density, computer basics, pH, light properties

Arithmetic: Order of operations, percentages, ratios, triangles/angles, algebra, perfect squares, speed/distance, discount problems

IQ: Tank-filling problems, averages, direction/distance, number series, coded operations, odd one out

2017

English: Synonyms, antonyms, prepositions, spellings, indirect narration, passive voice

General Knowledge: World rivers, Islamic history (Cave Hira, Masjid-e-Quba), Pakistani history, landmasses, landlocked countries, Allama Iqbal

General Science: Elements/symbols, blood pressure, solar system, human body (kidneys, blood vessels), lenses, citrus fruits, vitamin discovery

Arithmetic: Roots, partnership/ratio, wages/overtime problems, profit/loss, algebra (equations, sequences), LCM, sets, subsets

IQ: Rotating dot puzzles, handshake problems, grid/number patterns, coded ratios, word-letter grid puzzles, day/date problems

2018

English: Synonyms, antonyms, prepositions, spellings, narration conversion, passive voice

General Knowledge: Islamic history (Prophethood, Hazrat Ali's sword), Sindh culture, world capitals/landmarks, currencies, Zulfiqar Ali Bhutto's works

General Science: Solar system, milk composition, electric current, lenses, energy conversion, conductors, cardiovascular system, waves

Arithmetic: Percentages, algebra (equations, cubic roots), tank-filling, cube volume, HCF, decimals, factorization

IQ: Number series, handshake problems, classification, age problems, coded number patterns, geometric puzzles

2019

English: Synonyms, prepositions, spellings, sentence correction, error identification (longer/complex sentences)

General Knowledge: World geography (seas, oceans), Pakistani authors, Mughal history, Sindh (Mirpurkhas), Islamic history, capitals

General Science: Human body temperature, metals, railway physics, energy types, Newton's laws, cold-blooded animals, RADAR, density

Arithmetic: Irrational/rational numbers, trigonometric identities, logarithms, binary numbers, sets (De Morgan's Law), averages, ratios

IQ: Word-letter puzzles, shape patterns, handshake problems, race position logic, sock/probability problems, pill-timing logic

PET-2025 Sample Paper (Official Format)

English: Synonyms, antonyms, spellings, prepositions

General Science: Chemical formulas, SI units, geometry basics, computer hardware

General Knowledge: Currencies, Quran, Pakistani government structure, historical city names, national symbols, barrages

Mathematics: Percentages, decimals, unit conversion

IQ: Number series/patterns

4. Most Repeated Topics (Priority List)

These topics/patterns show the strongest recurrence across multiple exam sessions — prioritize these first.

1. Handshake problems (combinatorics) — appeared in 2014, 2015, 2017, 2018, 2019, sometimes with identical wording
2. Synonyms & Antonyms — present in every single year, largest recurring English category
3. Correct Spellings — appeared every year as a fixed sub-section
4. Prepositions (fill-in-the-blank) — recurring every year
5. Passive Voice conversion — recurring every year
6. Indirect/Direct Narration — recurring every year
7. Partnership/Ratio profit-sharing problems (e.g., Athar, Azhar, Mazhar business share) — repeated in 2014, 2017, 2018 with nearly the same structure
8. Profit/Loss percentage problems — repeated across 2013, 2014, 2017, 2018
9. Nested radical questions (e.g., $\sqrt{22 + \sqrt{6 + \sqrt{9}}}$) — appeared identically in 2015 and 2018
10. Sets (Union, Intersection, Subsets, De Morgan's Law) — repeated 2013, 2015, 2017, 2019
11. CNG full form (Compressed Natural Gas) — repeated in 2014 and 2018
12. Vitamin discoverer – Casimir Funk — repeated in 2017 and 2018
13. Normal blood pressure (120/80) — repeated in 2017 and 2018
14. Aorta = body's largest blood vessel — repeated in 2017 and 2018
15. Liver produces bile — repeated in 2013, 2014, 2017
16. Currency-based GK questions (Malaysia–Ringgit, Indonesia–Rupiah, Saudi Arabia–Riyal, China–Yuan) — recurring every year
17. Barrages of Pakistan (Kotri/Ghulam Muhammad Barrage, Sukkur Barrage) — repeated in 2013, 2017, 2018, and 2025 sample
18. Allama Iqbal's birth date (9th Nov 1877) and related facts — repeated in 2017 and 2018
19. Shah Abdul Latif Bhittai / Tambooro instrument — repeated in 2017 and 2018
20. Coded number pattern question "72496 is to 1315..." — appeared identically in 2017 and 2018
21. Sindh University Jamshoro being oldest university of Sindh — repeated in 2014 (similar "oldest university" fact in 2019 for Punjab)
22. Speed, distance, time word problems — repeated across 2015, 2017, 2019
23. Newton's Laws of Motion — repeated in 2015, 2019
24. Basic algebraic identities (a^2-b^2 , a^3+b^3 , factorization) — recurring every year

5. Verbatim Repeated Questions (As They Appear in Papers)

The exact wording and answers as found in the source papers, grouped by recurring theme.

1. Handshake Problems

[2014] *At a conference, 10 members shook hands with each other before and after the meeting. How many hand shakes occurred before?*

Answer: 45

[2017] *At a conference, 12 members shook hands with each other before and after the meeting. How many total member of hand shakes occurred?*

Answer: 132

[2018] *At a conference, 13 members shook hands with each other before and after the meeting. How many total number of handshakes occurred?*

Answer: 156

[2019] *At the end of a meeting 10 people shake hands with each other. How many handshakes will there be in total?*

Answer: 45

2. Partnership / Ratio Profit-Sharing

[2014] *Athar, Azhar and Mazhar opened a milk shop in the partnership. Share of profits is in the ratio of 1:2:3. If the profit is Rs.600, what is the Athar's share?*

Answer: Rs.100

[2017] *Athar, Azhar and Mazhar started a business in the partnership, share of profit is in the ratio of 2:3:4, if Azhar's share of profit is Rs.450, what is Athar's share?*

Answer: Rs.300

[2018] *Three brothers Athar, Azhar and Mazhar share 480 kg of rice. Azhar receives twice rice as much as Athar and Mazhar receives half rice as much as Azhar. How much rice does Azhar get?*

Answer: 240 kg

3. Two Apples / Three Mangoes Problem

[2017] *Two apples and three mangoes cost Rs.86, four apples and one mango cost Rs.112, what is the cost price of an apple?*

Answer: Rs.25

[2018] *Two apples and three mangoes cost Rs.86, four apples and one mango cost Rs.112. what is the cost of one apple?*

Answer: 25

4. Nested Radical

[2014] $\sqrt{1 + \sqrt{5 + \sqrt{16}}}$

Answer: 2

[2017] $\sqrt{22 + \sqrt{6 + \sqrt{9}}}$

Answer: 5

[2018] $\sqrt{22 + \sqrt{6 + \sqrt{9}}}$ is equal to?

Answer: 5

5. Sets (Union / Intersection / Subsets)

[2013] *If, $A = \{c, d, b, h\}$ and $B = \{m, p, f, h\}$ then $A \cup B = ?$*

Answer: None

[2015] *Which of the following is true? If, $A = \{1, 2\}$, $B = \{a, b\}$*

Answer: Equivalent set

[2017] *which of the following is a subset of $\{b, c, d\}$?*

Answer: Both a and b

[2018] *which of following is a subset of $\{b,c,d\}$*

Answer: Both a and b

[2019] *which of the following is true according to De-Morgan's Law?*

Answer: $(A \cup B)' = A' \cap B'$

6. CNG Full Form

[2014] *CNG stands for?*

Answer: Compressed Natural Gas

[2018] *CNG stands for?*

Answer: Compressed Natural Gas

7. Vitamin Discoverer

[2017] *The first person who found vitamins was?*

Answer: Casimir Funk

[2018] *The first person who found vitamins was?*

Answer: Casimir Funk

8. Normal Blood Pressure

[2017] *Normal blood pressure of healthy adult person is?*

Answer: 120/80

[2018] *Normal blood pressure of healthy adult person is?*

Answer: 120/80

9. Aorta

[2017] *The body's largest blood vessel is?*

Answer: Aorta

[2018] *The body's largest blood vessel is?*

Answer: Aorta

10. Liver Produces Bile

[2014] *Which part of Human body produces bile?*

Answer: Liver

[2015] *Which part of the human body produces bile?*

Answer: Liver

[2018] *Which part of body filter the blood coming from the digestive tract, before passing it to the rest of the body?*

Answer: Liver

11. Barrages

[2013] *The largest barrage of Pakistan is?*

Answer: Sukkur barrage

[2014] *Which barrage is also known as Ghulam Muhammad Barrage?*

Answer: Kotri Barrage

[2018] *Ghulam Muhammad Barrage is also known as?*

Answer: Kotri Barrage

[2025 Sample] *Which barrage is not located in Sindh?*

Answer: Chashma Barrage

12. Allama Iqbal's Birth Date

[2017] *Allama Muhammad Iqbal born on?*

Answer: 9th November, 1877

[2018] *Allama Muhammad Iqbal Born On?*

Answer: 9th November, 1877

13. Tambooro (Shah Abdul Latif Bhittai)

[2017] *Shah Abdul Latif Bhittai invented a Musical instrument called?*

Answer: Tambooro

[2018] *Shah Abdul Latif Bhittai invented a musical instrument called?*

Answer: Tambooro

14. Coded Number Pattern

[2017] *72496 is to 1315, 62134 is to 97 and 85316 is to 167, then 28439 is to?*

Answer: 1412

[2018] *72496 is to 1315, 62134 is to 97 and 85316 is to 167. Then 28439 is to?*

Answer: 1412

15. Oldest University

[2014] *The oldest University of Sindh province?*

Answer: Sindh University Jamshoro

[2019] *The oldest University of Pakistan is?*

Answer: University Of Punjab, Lahore